

Zakaria Khan

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EDUCATION

Wayne State University

Detroit, MI

Bachelor of Science: Physics, Chemistry ; Focus: Materials, Energy ; GPA: 3.9/4.0

May 2025

EXPERIENCE

Fulbright Scholar in Physics of Energy Materials

Aug 2025 – Present

U.S. Dept of State

Kuala Lumpur, WPKL, Malaysia

- Conducting Monte Carlo simulations, DFT calculations for electron transport, quantum mechanical effects in exotic energy materials: MoS2, topological insulators, 2D materials, perovskites
- Directing project guidelines, leading team of 2 graduate students, 4 Ph.D. scientists
- One of 14 grants in physics (2025) funded by U.S. Department of State's Fulbright program, valued at USD 25,000
- Work resulting in talk at FAST spintronics conference (Phuket, TH), 2 arXiv manuscripts (cond-matt)

Young Scientist in Interdisciplinary Energy

June 2025 – July 2025

Nobel Foundation, German Research Foundation

Lindau, Bavaria, Germany

- One of 20 undergraduates selected internationally
- Contributed to broad discussions with Nobel Laureates on energy science, diplomacy, global scientific challenges
- Received funding from DFG including travel, accommodation, attendance valued at USD 5,000.

Goldwater Scholar in Natural Science

June 2021 – April 2025

Wayne State University

Detroit, MI, USA

- Applied attosecond laser pulses to probe electron dynamics in perovskites, scintillators, other energy materials
- Wrote code for analysis of spectroscopic, spectral results
- Sought, received UROP funding, published, presented results several times across country GCURS, AGU, etc
- Awarded Barry Goldwater award: highest honor for undergraduate contributions to scientific research, valued at USD15,000 (1st WSU awardee, subsequently advised another WSU student who became 2nd)

Research Intern in Energy & Earth

May 2024 – Aug 2024

Department of Energy

Los Alamos, NM, USA

- Aided energy scientists in building 950 liter cloud chamber for fundamental energy science studies
- Configured, attached hardware to cloud chamber using swagelok fittings to be used for DOE grant studies
- Applied analytical instruments to quantify effects of mesoscale particle coagulation on energy processes, scattering
- Wrote code for pulling data off instruments, data analysis, visualization
- Wrote 40+ page report, delivered findings in talk to 100+ LANL staff scientists, trainees

PROJECTS

Q Dot Grid | C/C++, Monte Carlo, Docker, pandas, GIT

- Developed quantitative model for energy carrier transport in network of quantum dots
- Translated complex physical system into computational framework
- Collaborated with 2 Ph.D. students, contributing to model development, computational analysis, visualization
- Built, documented reproducible computational workflow, analyzed results to identify energy transport patterns

NRG Materials Tunnel | Python, DFT, Quantum ESPRESSO, HPC

- Analyzed electron transport in 2D materials to investigate potential for next-gen energy, electronic technologies
- Developed computational workflows to process, analyze large datasets, running parallel computing codes on HPC
- Developed Python workflows to automate computational analysis, process tunneling data from DFT calculations
- Applied quantitative modeling to connect material-level properties with broader questions surrounding energy tech

TECHNICAL SKILLS

Languages: Python, C/C++, Fortran, MATLAB, Bash, SQL, L^AT_EX

Scientific Computing: NumPy, SciPy, pandas, Matplotlib, Jupyter, MPI, OpenMP, CUDA

HPC Tools: Linux, Git, Docker, SLURM, GitHub, VS Code, CMake

Libraries: Quantum ESPRESSO, VASP, LAMMPS, DFT, Monte Carlo